A Overview of the autism spectrum disorder run (436 input genes)

Input genes
431 / 436
genes recognized;
5 unresolved

Pathway hypotheses
100
generated in total, across two prompt passes

Validated pathways
60
significant after enrichment testing in five databases

Cell-type context
60 / 60
validated pathways carry a cell- or tissue-context statement

Validated pathways per database

GO:BP	21
GO:MF	12
GO:CC	12
KEGG	9
REAC	6

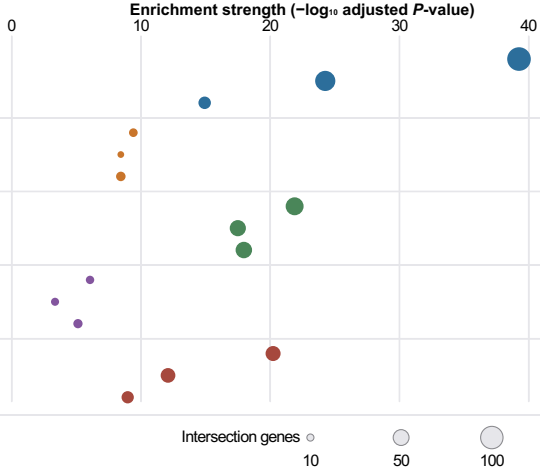
Cell-type context across the 60 validated pathways

Cortical neurons	44 of 60 (73%)
Hippocampal neurons	35 of 60 (58%)
Pyramidal neurons	25 of 60 (42%)
Excitatory neurons	18 of 60 (30%)
Inhibitory neurons	6 of 60 (10%)

B Pathway enrichment overview

Three highest biologically ranked pathways per database

Rank		Pathway	Adjusted *P*-value
#1	**GO:BP**	chromatin organization	5.51×10^{-40}
#2		neuron projection development	5.40×10^{-25}
#3		regulation of synaptic plasticity	1.15×10^{-15}
#1	**GO:MF**	histone methyltransferase activity	3.81×10^{-10}
#2		ATP-dependent chromatin remodeler activity	3.54×10^{-9}
#3		chromatin DNA binding	3.54×10^{-9}
#1	**GO:CC**	postsynapse	1.26×10^{-22}
#2		dendrite	3.14×10^{-18}
#3		axon	1.07×10^{-18}
#1	**KEGG**	Long-term potentiation	8.53×10^{-7}
#2		Glutamatergic synapse	4.35×10^{-4}
#3		Dopaminergic synapse	7.35×10^{-6}
#1	**REAC**	Chromatin organization	5.87×10^{-21}
#2		Neuronal System	7.89×10^{-13}
#3		Transmission across Chemical Synapses	1.04×10^{-9}



Point area, genes shared with the input list; point colour, database.

C Interpretation of a representative high-ranking pathway

GO:BP · biological rank #2
neuron projection development
GO:0031175

1 Statistical evidence
FDR-adjusted *P*-value = 5.40×10^{-25}
overlap 81 of 1,010 annotated genes

2 Key intersection genes
81 genes, including four clusters; principal drivers *CNTNAP2*, *FMR1* and *PTEN*

3 Cell and tissue context
Developing cortical and hippocampal projection neurons and their axons and dendrites

Interpretation

Neuron projection development (GO:0031175) ranked second among GO:BP pathways. Its 81-gene intersection contains four clusters: synaptic scaffolding and signalling (*CAMK2A*/*B*, *NLGN2*), transcriptional and chromatin regulation of projection identity (*ADNP*, *FOXG1*, *MECP2*, *PAX6*), guidance and adhesion (*CNTNAP2*, *DSCAM*, *RELN*), and vesicle trafficking (*KIF1A*, *KIF5C*, *NRXN1*, *DLG4*), so upstream transcriptional regulators and downstream synaptic effectors sit in one process. *CNTNAP2*, *FMR1* and *PTEN* carry the strongest human genetic evidence and each controls neurite arborisation, spine maturation or neuronal growth. *Chd8*-haploinsufficient mouse cortex [1] and single-cell methylation profiling of ASD prefrontal cortex [2] link transcriptional dysregulation during cortical development to neurite growth, and *Clcn4*-null mice show the corresponding dendritic deficits [3]. The cell-context statement places the signal in developing cortical and hippocampal projection neurons [4].

PubMed records retrieved by the platform

PMID 40967226 Cell-type-specific dysregulation of gene expression due to Chd8 haploinsufficiency… [1]
PMID 42320469 Single-cell profiling of DNA methylation in autism spectrum disorder prefrontal cortex… [2]
PMID 39863599 Developmental deficits, synapse and dendritic abnormalities in a Clcn4 KO autism mice model… [3]
PMID 34971749 Length impairments of the axon initial segment in rodent models of ADHD and ASD [4]